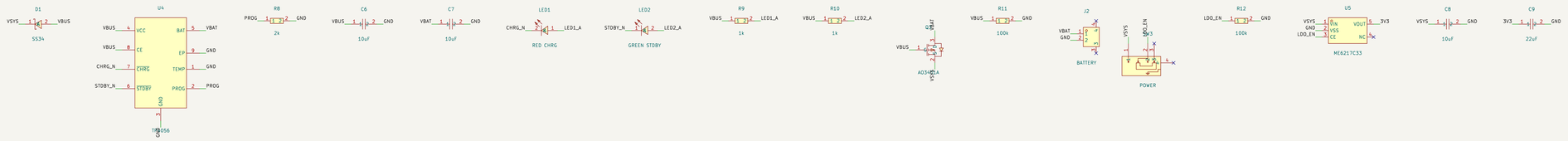
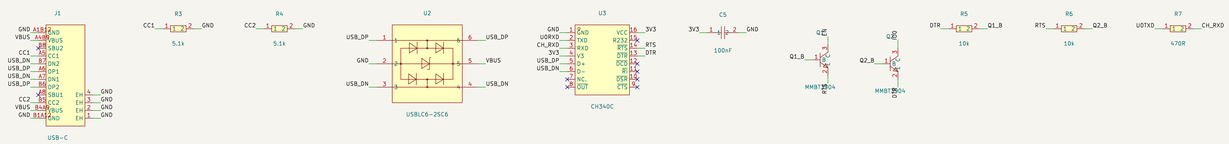
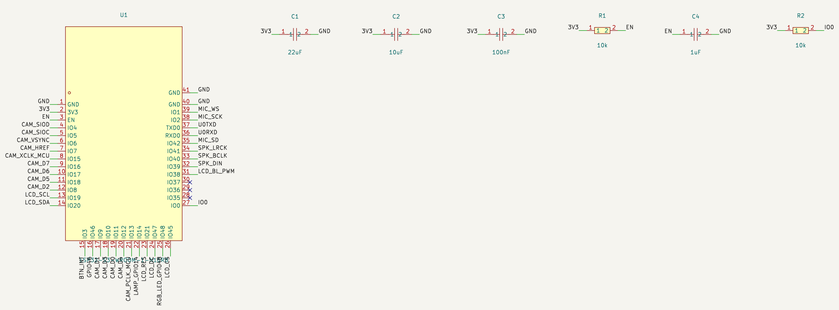
## POWER: USB-C 5V / TP4056 charger / power-path / 3V3 LDO



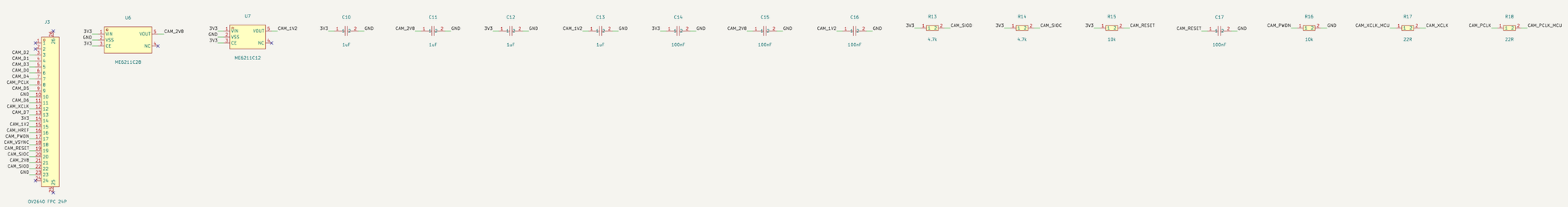

## USB-UART CH340C + auto-download (DevKitC style)



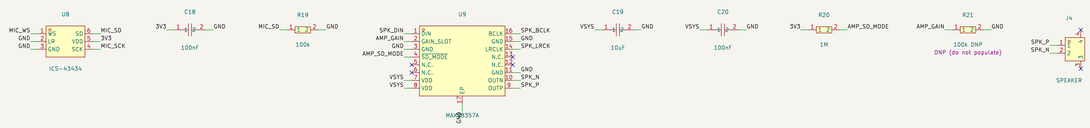

## ESP32-S3-WROOM-1-N16R8 core



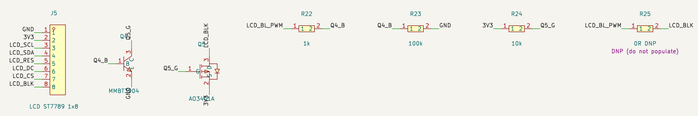

## CAMERA OV2640 24P DVP (xiaozhi bread-compact-wifi-s3cam)




## AUDIO: ICS-43434 I2S mic + MAX98357A I2S amp

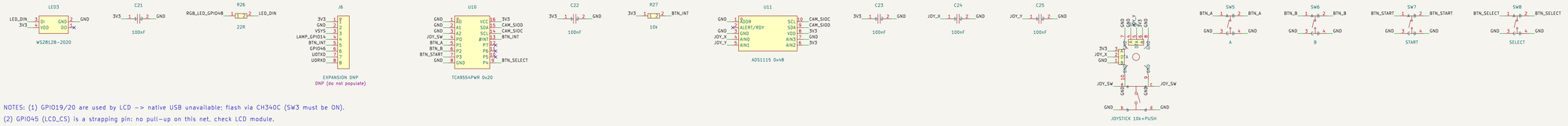



## LCD ST7789 240x320 SPI + backlight




## WS2812 status LED + expansion header

## GAME CONTROLS (v3): ADS1115 @0x48 + TCA9554 @0x20 on SCCB bus (GPIO4/5), INT->GPIO3, analog joystick + A/B/START/SELECT




NOTES: (1) GPIO19/20 are used by LCD -> native USB unavailable; flash via CH340C (SW3 must be ON).
(2) GPIO45 (LCD_CS) is a strapping pin: no pull-up on this net, check LCD module.
(3) J3 FPC pin order = ESP32-CAM style (pin1=Y0). Verify against your OV2640 module + connector contact side.
(4) R21, R25, J6 = DNP. Battery J2 polarity: pin1 = +, verify your battery lead.
(5) v3: U10 TCA9554 (0x20) + U11 ADS1115 (0x48) share camera SCCB (GPIO4 SDA / GPIO5 SCL). INT -> GPIO3 (R27 pull-up). SW4 joystick X->AIN0, Y->AIN1, AIN2=3V3 ref, push->P0.
(6) v3: SW4 Alps RKJXV122400R is THT; drill sizes tightened to Alps mounting-hole drawing. Official xiaozhi firmware ignores joystick/buttons.

Symbols/footprints: LCSC/EasyEDA via easyeda2kicad. GPIO45 = LCD CS: NO pull-up (strapping).
Auto-generated from BOM/pin map. NOT built or tested. Check every footprint & FPC orientation.

Maker Bu Wan Ji Hua

Sheet: /
File: ESP32S3_AI_Chatbot.kicad_sch

**Title: ESP32-S3 AI Chatbot (xiaozhi bread-compact-wifi-s3cam pinout)**

| | |
|---|---|
| Size: A1 | Date: 2026-10-06 |
| KiCad E.D.A. 9.0.2 | Rev: 0.3 (v3 analog joystick) |
| | Id: 1/1 |